ECO-3500
Evaluating the Impact of BBBP on Female Educational Outcomes in India

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Overview of BBBP

- **Context:** Pre-2015 India experienced a persistent decline in the sex ratio at birth, pointing to strong son preference and gender-biased investments (Census 2011).
- **BBBP (2015):** Aims to "save the girl child, educate the girl child" through awareness campaigns, community mobilisation, and behavioural messaging, rather than material subsidies.
- **Design**
  - Multi-sectoral strategy: health, education, PC-PNDT monitoring.
  - Heavy emphasis on mass media campaigns
- **Rollout Timeline**
  - Phase I (2015–16): 100 "gender-critical" districts
  - Phase II (2016–17): +61 districts
  - Nationwide (2018): 244 multisectoral + 235 media-only districts

Our Research Question

- **Core Question:** Does the Beti Bachao, Beti Padhao (BBBP) scheme causally improve female educational outcomes in India?
- **Why This Matters:**
  - Awareness-driven interventions are high visibility but costly $\rightarrow$ need evidence of real impact.
  - Improving girls' education is central to long-run development and labour market participation.
  - Understanding whether behavioural messaging alone can shift schooling outcomes informs policy design, especially for gender equity.

Theoretical Framework

- In India, major supply constraints in schooling have largely been addressed; the persistent barriers to girls' education are demand-side -parental beliefs, norms, and perceived returns.
- BBBP targets these demand-side constraints by attempting to change gender norms and increase the value placed on educating daughters.
- Therefore, if BBBP succeeds in shifting beliefs, we should see measurable improvements in girls' schooling and learning outcomes.

Literature and Gaps

- **Prior Findings**
  - BBBP significantly increased awareness among target groups (NCAER).
  - Some evidence of improved sex ratio at birth, especially in Haryana (Gupta et al.).
  - Several critiques include inefficient fund allocation, weak monitoring and data gaps, heavy emphasis on publicity over substantive health/education interventions
- **What's Missing:**
  - Existing research focuses almost entirely on demographic indicators (SRB), not education.
  - Most studies evaluate the programme only in Haryana, and many were government-commissioned, limiting independent scrutiny.

Our Contribution

- **Data:** We use child and HH level data from **ASER**, a nationally representative rural survey measuring enrollment and foundational learning.
  - Data available for years 2007-14, 2016, 2018, 2022, 2024.
  - Effectively in data, **post years for Phase I and II are the same (2016+)**.
- **Methodology: DiD** with two-way fixed effects.
- **Goal:** Identify the causal impact of BBBP's *awareness-driven* rollout on:
  - Girls' reading and math scores.
  - School dropout rates (Class 5 and 8).
- **Value:** assessed whether behavioural/awareness policy tools alone can translate into measurable improvements in schooling outcomes.

Specification A: Learning Outcomes

Objective: Estimate impact on foundational Reading and Math.

- **Sample:** Girls in Classes 1, 2, and 3 (separate regressions).
- **Unit of Analysis:** Child (i) in district (d), class (c), year (t).

Estimating Equation:

$$\begin{aligned} Y_{idct} = \alpha + \beta_1(\text{PhaseI}_d \times \text{Post}_t) + \beta_2(\text{PhaseII}_d \times \text{Post}_t) \\ + X'_{idct}\theta + \gamma_d + \delta_t + \varepsilon_{idct} \end{aligned} \tag{1}$$

- Y_{idct}: Standardized Math or Reading proficiency score.
- PhaseI/II_d: Indicators for treatment cohorts.
- Post_t: Indicator for years ≥ 2015.
- γ_d, δ_t: District and Year fixed effects.
- X'_{idt}: Controls (Child, Parental, Wealth, Village)
- *Standard errors clustered at the district level.*

Specification B: Dropout Likelihood

Objective: Analyze exit rates at key structural transition points.

- **Sample:** Girls reported as "out of school".
- **Focus:** Separate regressions for dropout by **Class 5** and **8**.

$$\text{Dropout}_{idt} = \alpha + \beta_1(\text{PhaseI}_d \times \text{Post}_t) + \beta_2(\text{PhaseII}_d \times \text{Post}_t) + X'_{idt}\theta + \gamma_d + \delta_t + \varepsilon_{idt} \tag{2}$$

- Dropout_{idt}: $= 1$ if girl dropped out in specified class.

Vector of Controls (X'_{idt}):

Child & Parental:

- Tuition attendance
- Private school indicator
- Mother & Father education

Wealth & Village:

- Pucca house, Electricity, TV
- Pucca road availability
- Gov. School access (Prim/Mid/Sec)

Descriptive Evidence: Target Groups

Summary Statistics of Scores by Class for Girls

Variable	Obs	Mean	Std. Dev.
Class 1			
Reading Score	299,231	1.989	1.009
Math Score	297,680	1.988	0.906
Class 2			
Reading Score	286,453	2.687	1.185
Math Score	285,309	2.560	0.977
Class 3			
Reading Score	286,239	3.278	1.261
Math Score	285,290	3.026	1.050

Learning Gap: Girls' test scores are far below grade-appropriate levels. Even by Class 3, average reading and math scores remain around 3/5. We focus on learning in early primary.

Distribution of School Dropouts by Class for Girls

Class Dropped Out	Frequency	Percent (%)
Class 0 (Pre-primary)	101	0.12
Class 1	2,346	2.85
Class 2	4,437	5.40
Class 3	5,702	6.94
Class 4	5,782	7.04
Class 5	17,124	20.84
Class 6	6,748	8.21
Class 7	8,195	9.97
Class 8	16,933	20.61
Class 9	6,946	8.45
Class 10	7,072	8.61
Class 11	389	0.47
Class 12	393	0.48
Other/Unknown (99)	5	0.01
Total	**82,173**	**100.00**

Dropout at Transition Grades: Sharp dropout spikes at Class 5 and Class 8 suggest girls leave school at key transition points. We focus on these grades for dropout likelihood.

Findings: Math and Reading Scores

	Math Score			Reading Score		
	Class 1	Class 2	Class 3	Class 1	Class 2	Class 3
Phase I × Post						
	0.0209	0.0125	0.0156	0.0644*	0.1035***	0.1053***
	(0.0282)	(0.0305)	(0.0322)	(0.0340)	(0.0349)	(0.0366)
Phase II × Post						
	0.0908***	0.0847**	0.0913*	0.1207***	0.2061***	0.1908***
	(0.0281)	(0.0393)	(0.0467)	(0.0315)	(0.0411)	(0.0502)
Observations	297,680	285,309	285,290	299,231	286,453	286,239
Clusters (district)	649	649	649	649	649	649
R^2	0.229	0.206	0.207	0.199	0.195	0.190
FE: District, Year	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Each column reports a separate regression for girls in Classes 1–3. All specifications include district and year fixed effects and are clustered at the district level. Controls include: *takes_tuition, private_school, mom_edu, dad_edu, pucca_house, has_tv, has_elec, has_road, has_prim_school, has_mid_school, has_sec_school.* Math and reading outcomes are standardized test scores.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{\dagger}p < 0.10$

Figure 1: Impact of Phase I and Phase II on Girls' Math and Reading Scores

Findings: Dropout Likelihood

	Dropout Class 5	Dropout Class 8
Phase I × Post		
	-0.01126	-0.00158
	(0.01138)	(0.01411)
Phase II × Post		
	0.00677	-0.00763
	(0.01366)	(0.01654)
Observations	98,099	98,099
Clusters (district)	644	644
R^2	0.0850	0.0957
Within R^2	0.0104	0.0059
FE: District, Year	Yes	Yes
Controls Included	Yes	Yes

Notes: Each column reports a separate HDFE regression with district and year fixed effects. All standard errors are clustered at the district level. Controls include: *takes_tuition, mom_edu, dad_edu, pucca_house, has_tv, has_elec, has_road, has_prim_school, has_mid_school, has_sec_school.*
Dropout outcomes are binary indicators for girls who report being out-of-school. The sample is restricted to *child_gender* = 2 and *oos_dropout* = 1.

Figure 2: Impact of Phase I and Phase II on Girls' Dropout Likelihood

Key Results: Impact of BBBP on Girls' Education

1. Learning Outcomes (Classes 1–3)

- **Phase I** $\times$ **Post:** No effect on Math; small but significant gains in Reading (esp. Class 2–3).
- **Phase II** $\times$ **Post:** Consistently positive significant effects on **Math** ($+0.08$–0.09 pts) and **Reading** ($+0.12$–0.21 pts).
- *Mechanism:* Stronger improvements in reading suggest parents can more easily support literacy at home than math.

2. Dropout Outcomes (Classes 5 & 8)

- **Result:** No significant effect of BBBP (Phase I or II) on dropout rates at key transition grades.
- *Implication:* Indicates deeper **structural barriers** (economic constraints, safety, access to secondary school) remain unchanged.

Conclusion: BBBP improved early learning slightly, but did not reduce dropout. Awareness campaigns shift early learning somewhat but cannot overcome structural challenges without complementary resource-intensive interventions.

Policy Implications

- **Awareness can shift early learning**, especially reading, but its effects are modest and uneven across subjects.
- **Combine messaging with quality improvements** (teacher support, remedial instruction, classroom resources) to strengthen learning gains.
- **Program design matters** - stronger Phase II effects suggest that effective components (media, community work, multi-sectoral efforts) should be identified and scaled.
- **Dropout needs structural support** (safe transport, financial incentives, secondary-school access), as awareness alone does not change continuation at higher grades.

Limitations

- **Non-random rollout:** Phase I and II districts were specifically chosen because they had low or declining CSR, which may create selection bias.
- **Parallel trends:** We assume that without BBBP, both treated and untreated districts would have followed similar trends over time but we haven't demonstrated that this actually happened.
- **Spillovers:** Mass-media campaigns may reach neighbouring districts, creating spillovers that make treated and untreated districts less comparable.